

SMRTTECH 4AI3 - Artificial Intelligence and Machine Learning

Lab 3 - Data Pre-processing & Classification

Instructor: Siqi Zhao

Due: Midnight 11:59PM, day before your next lab session

Please read and follow these instructions:

- Submit your lab on Avenue to Learn (A2L) before the due date (midnight of the day before your next lab session). The submission will be closed after the deadline.
- Do your lab exercises on your own, i.e., do not copy. If two lab reports (or some parts of them) are found to be copies of each other, then BOTH shall receive 0 mark.
- You may use Google Colab or any other Python programming platform of your own choice to run your programs.
- Please read the questions carefully and use AI / machine learning libraries only when specially required as part of the question. Also add comments in your code where ever necessary.
- Submit your working python source files (or Jupyter notebooks) in zipped or source code format.

1 Explore the Hamilton Waterfalls Dataset

The Waterfalls dataset is provided by the Open Hamilton, a project designed to enable the community to explore and download the city data. This dataset contains information on publicly accessible waterfalls in Hamilton, on both public and private property.

2 Exercises (20 Points)

1. Download this dataset **City_Waterfalls.csv** from **A2L** and upload it in you Colab directory. Take a look at this dataset and make sure the following headers exist: **(2 Points)**

- | | |
|----------------------|-------------------------------------|
| • LONGITUDE | # longitude of the waterfall |
| • LATITUDE | # latitude of the waterfall |
| • NAME | # the name of the waterfall |
| • ACCESS FROM | # entry point of the waterfall |
| • HEIGHT_IN_M | # height of the waterfall in meters |
| • WIDTH_IN_M | # width of the waterfall in meters |

- **OWNERSHIP** # either publicly or privately owned
- **RANKING** # ranking based on survey
- **TYPE** # if the waterfall is a cascade

2. Import the related packages: **(1 Point)**
3. Use `read_csv()` method from Pandas to load the dataset into a Pandas DataFrame named `waterfalls`. After that, use `.head()` to check the first five rows of the data points. **(1 Point)**
4. Use `.info()` to check the data type of each feature in the dataset. After that, use the `.describe()` method to see a statistical summary of the numerical features in the dataset **(1 Point)**
5. If we would like to know which community in Hamilton has the most waterfalls, how do we check it? (Hint: We can use `.value_counts()` method on the `COMMUNITY` feature.) **(1 Point)**
6. Use `.value_counts()` on the `RANKING` feature to check the distribution of three different rankings on the waterfalls. **(1 Point)**
7. Let's select certain features in the original DataFrame: `NAME`, `LONGITUDE`, `LATITUDE`, `HEIGHT_IN_M`, `WIDTH_IN_M`, `OWNERSHIP`. **(1 Point)**
8. Generate a scatter plot based on the longitude and latitude of the waterfalls. **(1 Point)**
9. Generate a histogram plot `HEIGHT_IN_M` and `WIDTH_IN_M`. **(1 Point)**
10. Assume we want to predict the **OWNERSHIP** of the waterfall by the height and width of the waterfall, select the `HEIGHT_IN_M` and `WIDTH_IN_M` as inputs (features), and the `OWNERSHIP` as output (label). **(1 Point)**
11. Encode the categorical column 'OWNERSHIP' as numerical values **(1 Point)**
12. Split the dataset into training set and test set by the ratio 4:1. **(1 Point)**
13. Build Logistic Regression Model by using sklearn (`max_iter=1000`) **(1 Point)**
14. Train the logistic regression model by training set **(1 Point)**
15. Predict the testing set ownership **(1 Point)**
16. Evaluate the model using testing set, calculate the Accuracy, confusion Matrix, Precision, recall and F1-Score **(4 Points)**